



DESIGN AND ANALYSIS OF ALGORITHMS

UE19CS251

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DESIGN AND ANALYSIS OF ALGORITHMS

Merge Sort

Major Slides Content: Anany Levitin

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Merge Sort - Idea

- Split array $A[0..n-1]$ into about equal halves and make copies of each half in arrays B and C
- Sort arrays B and C recursively
- Merge sorted arrays B and C into array A as follows:
 - Repeat the following until no elements remain in one of the arrays:
 - compare the first elements in the remaining unprocessed portions of the arrays
 - copy the smaller of the two into A, while incrementing the index indicating the unprocessed portion of that array

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Merge Sort - Algorithm



```
ALGORITHM Mergesort ( A[ 0 .. n- 1] )
// Sorts array A[ 0 .. n- 1] by recursive
  mergesort
// Input: An array A[ 0 .. n- 1] of orderable
  elements
// Output: Array A[ 0 .. n- 1] sorted in non
  decreasing order
if n > 1
  copy A[ 0 ..  $\lfloor n/2 \rfloor - 1$  ] to B[ 0 ..
    - 1]
  copy A[  $\lfloor n/2 \rfloor$  .. n - 1 ] to C[ 0 ..
    - 1]
  Mergesort ( B[ 0 ..  $\lfloor n/2 \rfloor - 1$  ] )
  Mergesort ( C[ 0 ..  $\lfloor n/2 \rfloor - 1$  ] )
  Merge( B, C, A)
```

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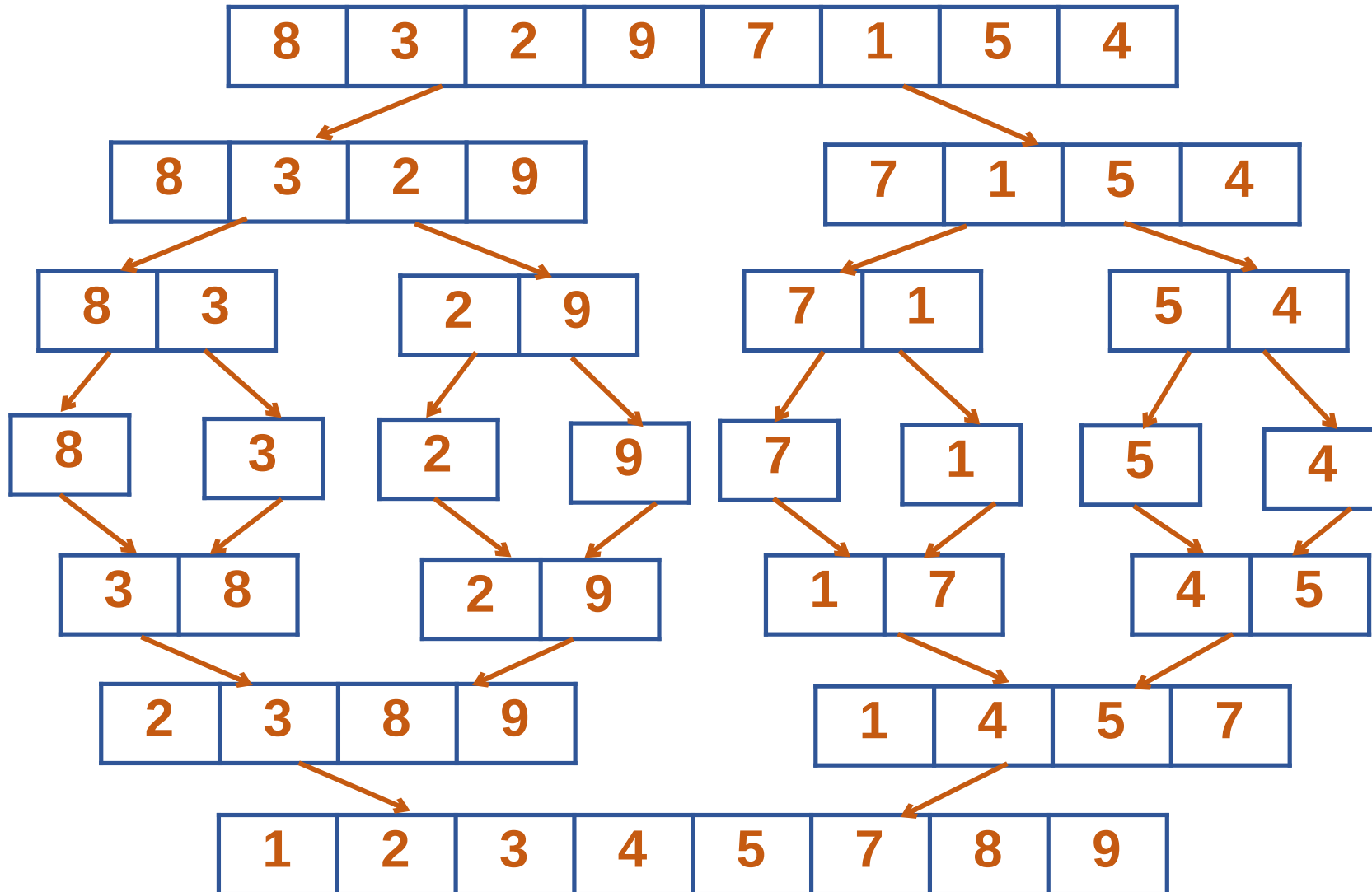
Merge Sort - Algorithm



```
ALGORITHM Merge( B[ 0 .. p- 1], C[ 0 .. q -
    1], A[ 0 .. p + q - 1] )
// Merges two sorted arrays into one sorted
  array
// Input: Arrays B[ 0 .. p - 1] and C[ 0 .. q -
    1] both sorted
// Output: Sorted array A[ 0 .. p + q - 1] of
  the elements of B and C
i ← 0; j ← 0; k ← 0
while i < p and j < q do
  if B[i] ≤ C[j]
    A[k] ← B[i]; i ← i + 1
  else A[k] ← C[j]; j ← j + 1
  k ← k + 1
if i = p
  for j = 0 to q - 1
    A[k + j] ← C[j]
```

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Merge Sort - Example



- Assuming for simplicity that n is a power of 2, the recurrence relation for the number of key comparisons $C(n)$ is:

$$C(n) = 2C(n/2) + C_{\text{merge}}(n) \quad [\text{for } n > 1], \quad C(1) = 0$$

- The number of key comparisons performed during the merging stage in the worst case is:

$$C_{\text{merge}}(n) = n - 1$$

- Using the above equation:

$$C_{\text{worst}}(n) = 2C_{\text{worst}}(n/2) + n - 1 \quad [\text{for } n > 1],$$

$$C_{\text{worst}}(1) = 0$$



THANK YOU

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